

legalvrp V0 — Model validation

Assumptions, variables, constraints and method of the daily MILP

For review and sign-off before implementation

How to use this document. Every assumption (§2) and every constraint (§6) has a box ☐. Tick it if you agree; write a note next to it if not. Anything you change goes back into `PROJECT_BRIEF.md` §6 and `docs/MODEL.md` before code is written. §11 lists the decisions that are genuinely yours to make.

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1 The problem in five lines

A regional distributor delivers pallets from one depot with rigid trucks over 3.5 t. Each afternoon it plans the next day: which driver serves which order, in what sequence, where the legal break is taken, and which orders are postponed. Plans must respect truck capacity, receiving windows, access and equipment, EU driving-time rules, EU and French working-time rules, shifts, and the hours each driver has already worked this week. V0 plans one week, Monday to Friday, as five daily MILPs solved exactly with Gurobi, each passing the drivers' weekly state to the next.

	tiny	small (main)	scale
Days	1	5 (Mon–Fri, rolling)	1
Drivers	2	4 (3 full-time, 1 part-time)	$\lceil n/5 \rceil$
Orders per day	6–8	15–20	$n \in \{8, \dots, 40\}$
Purpose	tests, CI	the V0 result	complexity measurement

2 Assumptions

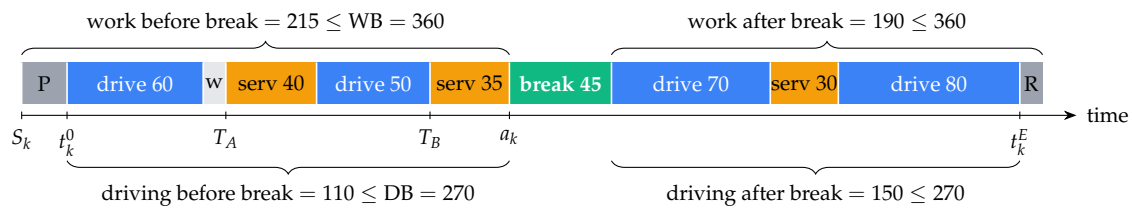
Every assumption below is *conservative*: if it is wrong, the model may reject a legal plan, but it never accepts an illegal one — except where stated.

#	Assumption	Why, and what it costs	OK
<i>Scope</i>			
A1	One depot; every duty starts and ends there the same day	Regional distribution. Excludes long-haul nights out	☐
A2	One driver drives one fixed truck all week	Avoids driver–truck swaps	☐
A3	Travel times deterministic, time-independent, integer minutes	No congestion model in V0	☐
A4	Service times deterministic ($\sigma = 0$)	ML layer comes later; the field already exists	☐
A5	Orders of day d are known at $d - 1$; no re-planning during the day	Matches the D–1 dispatching process	☐
<i>Legal interpretation</i>			
A6	At most one break per duty, 45 min, un-split	Enough for both break rules because daily driving $\leq 9\text{ h} = 2 \times 4\text{ h}30$ and daily work $\leq 12\text{ h} = 2 \times 6\text{ h}$. Forgoes split breaks	☐
A7	The break is taken at a customer, right after service	Makes the model linear. Forgoes en-route breaks; harmless while no leg exceeds 4 h30	☐
A8	One break resets both counters (driving and work)	How drivers behave; to confirm against official guidance	☐
A9	Waiting at a customer counts as work , never as a break	Conservative for the 6 h rule	☐
A10	45 min of break whenever work exceeds 6 h	The law asks only 30 min between 6 and 9 h; slight over-protection	☐
A11	Daily driving cap is 9 h; the twice-a-week 10 h extension is disabled	Simplicity; it is a per-driver weekly counter to add later	☐

#	Assumption	Why, and what it costs	OK
A12	Same shift start every day for a given driver	Makes the 11 h daily rest automatic (on duty ≤ 12 h45, so off duty ≥ 11 h15); still checked	☐
A13	<i>Temps de service</i> = everything on duty except the break: driving, service, waiting, depot prep P and close R	Follows the official definition (driving + other work + waiting)	☐
<i>Economics</i>			
A14	A salaried driver's hours below the contract threshold carry only a tie-breaker cost (0.01 €/min)	They are already paid; the real levers are km, extra hours, temps and postponements	☐
A15	Every order may be postponed to the next day at a penalty p_i ; Friday leftovers are reported unserved	Keeps every day feasible; see §6.8	☐
A16	Monetary values are illustrative	Calibrate from CNR indices if you want realism	☐

3 One duty, drawn

The figure fixes the meaning of the time quantities the constraints use. Numbers are illustrative.



Definitions used everywhere. *Driving* = sum of travel times. *Work* = elapsed time on duty excluding the break (waiting included). *Temps de service* θ_k = from S_k (start of prep) to $t_k^E + R$ (end of close), minus the break: here $450 - 45 = 405$ minutes. The break sits right after a service (A7); the 45 minutes delay the next departure.

4 Notation and parameters

Symbol	Meaning	Value / unit
C	orders of the day (including those postponed from yesterday)	
$0, E$	depot departure node, depot return node	
K	drivers working that day	
C_k	orders compatible with driver k (truck size vs access, tail-lift, pallets $\leq$ capacity)	
A_k	arcs of driver k : $(0, j), (i, j)$ with $i \neq j, (i, E)$, and $(0, E)$ = empty route	
τ_{ij}, d_{ij}	travel time, distance (asymmetric allowed)	min, km
$[e_i, l_i]$	window for the <i>start</i> of service	min from midnight
s_i, q_i	service duration, pallets	min, pallets
Q_k	truck capacity	pallets
S_k, F_k	shift start, latest end of duty	min
P, R	depot prep before departure, depot close after return	20, 10 min
DB, WB	max driving / work between breaks	270, 360 min
BR	break length	45 min
DD, DS	max driving / <i>temps de service</i> per day	540, 720 min
WD	max driving per week	3360 min
W_k, V_k	service / driving minutes already worked this week (state)	min
$H_k^{\text{thr}}, H_k^{\text{max}}$	weekly threshold for extra pay, weekly service cap	e.g. 2340, 3120 min
c^{km}	cost per km	1.60 €/km (illustr.)
$c_k^{\text{reg}}, c_k^{\text{ext}}$	cost per regular / extra minute	0.01, 0.45 €/min (illustr.)
c_k^{fix}	fixed cost if the driver works (temp agency day fee)	€
p_i	postponement penalty	200 €/order (illustr.)

Arc pruning (preprocessing). Arc (i, j) is not created if $e_i + s_i + \tau_{ij} > l_j$: even with no waiting and no break, j would be reached too late. Incompatible (driver, order) pairs create no variables at all.

5 Decision variables

Code	Symbol	Domain	Meaning	OK
$x[i, j, k]$	x_{ijk}	$\{0, 1\}$	driver k travels arc (i, j)	☐
$u[i]$	u_i	$\{0, 1\}$	order i is postponed	☐
$T[i, k]$	T_{ik}	$[e_i, l_i]$	service start at i (meaningful if k visits i)	☐
$D[i, k]$	D_{ik}	$[0, DD]$	driving accumulated on arrival at i	☐
$y[i, k]$	y_{ik}	$\{0, 1\}$	k takes the break right after serving i	☐
$t0[k]$	t_k^0	$[S_k + P, F_k]$	departure from the depot	☐
$tE[k]$	t_k^E	$[S_k + P, F_k - R]$	return to the depot	☐
$a[k]$	a_k	$[S_k, F_k]$	start of the break	☐
$b[k]$	b_k	$[0, DD]$	driving accumulated when the break starts	☐
$svc[k]$	θ_k	$[0, DS]$	<i>temps de service</i> today	☐
$ext[k]$	o_k	≥ 0	weekly minutes above the threshold, after today	☐

Shorthand expressions (not variables): $\text{visit}_{ik} = \sum_j x_{jik}$ (does k serve i), $\text{used}_k = 1 - x_{0Ek}$, $\text{drive}_k = \sum_{(i,j)} \tau_{ij} x_{ijk}$, $\text{brk}_k = \sum_i y_{ik}$ (0 or 1).

6 Objective and constraints

6.1 Objective

$$\min \sum_{k \in K} \left[\underbrace{c^{\text{km}} \sum_{(i,j) \in A_k} d_{ij} x_{ijk}}_{\text{distance}} + \underbrace{c_k^{\text{reg}} \theta_k}_{\text{tie-breaker}} + \underbrace{c_k^{\text{ext}} o_k}_{\text{extra hours}} + \underbrace{c_k^{\text{fix}} \text{used}_k}_{\text{temp day fee}} \right] + \sum_{i \in C} \underbrace{p_i u_i}_{\text{postponement}} \quad \square$$

Cost of tomorrow. Because o_k counts the whole week's minutes above the threshold, a driver who was already above it before today contributes a constant; it does not change which plan is optimal. The KPI reports only today's increment.

6.2 Covering and routing

C1 cover $\square \quad \sum_{k: i \in C_k} \text{visit}_{ik} + u_i = 1 \quad \forall i \in C$

Each order is served by exactly one compatible driver, or postponed.

C2 leave $\square \quad \sum_j x_{0jk} = 1 \quad \forall k$

Each driver has exactly one route; an idle driver takes the empty arc $(0, E)$.

C3 flow $\square \quad \sum_j x_{jik} = \sum_j x_{ijk} \quad \forall k, i \in C_k$

A driver who arrives at a customer leaves it.

C4 capacity $\square \quad \sum_i q_i \text{visit}_{ik} \leq Q_k \quad \forall k$

Delivery only, loaded once at the depot: the total load is all that matters.

6.3 Time

C5 time from depot $\square \quad T_{jk} \geq t_k^0 + \tau_{0j} - M_{jk}^0(1 - x_{0jk})$

C6 time between customers $\square \quad T_{jk} \geq T_{ik} + s_i + \text{BR } y_{ik} + \tau_{ij} - M_{ij}(1 - x_{ijk})$

C7 time back to depot $\square \quad t_k^E \geq T_{ik} + s_i + \text{BR } y_{ik} + \tau_{iE} - M_{ik}^E(1 - x_{iEk}), \quad \text{and} \quad t_k^E \geq t_k^0$

If k uses an arc, the next service cannot start before the previous one ends, the break (if taken there) is over, and the travel is done. Waiting is allowed ($\geq$). The windows are the variable bounds of T_{ik} .

Why this form. Because every $s_i + \tau_{ij} > 0$, these constraints also forbid subtours (time cannot go round in a circle), so no separate subtour elimination is needed.

6.4 Driving accumulation

C8 from depot $\square \quad \tau_{0j} - DD(1 - x_{0jk}) \leq D_{jk} \leq \tau_{0j} + DD(1 - x_{0jk})$

C9 between customers $\square \quad D_{ik} + \tau_{ij} - (DD + \tau_{ij})(1 - x_{ijk}) \leq D_{jk} \leq D_{ik} + \tau_{ij} + DD(1 - x_{ijk})$

D_{jk} is *exactly* the driving done before reaching j on k 's route.

Why this form. Both sides are needed. With the lower side only, the solver may *inflate* D at the break node, so the driving *after* the break looks shorter than it is. §9 shows a concrete plan that this error accepts illegally. The lower big-M is $DD + \tau_{ij}$, not DD : otherwise a heavily-driven node could wrongly constrain a successor through an unused arc.

6.5 The break

C10 where $\square \quad y_{ik} \leq \text{visit}_{ik} \quad \forall i; \quad \text{brk}_k \leq 1$

A break only at a customer the driver actually serves; at most one break.

C11 when $\square \quad T_{ik} + s_i - M_{ik}^a(1 - y_{ik}) \leq a_k \leq T_{ik} + s_i + M_{ik}^a(1 - y_{ik})$

C12 driving at the break $\square \quad D_{ik} - DD(1 - y_{ik}) \leq b_k \leq D_{ik} + DD(1 - y_{ik}); \quad b_k \leq DD \text{brk}_k$

If the break is after i , then a_k is the end of service at i and b_k is the driving done so far. Without a break, $b_k = 0$.

6.6 Legal limits within the day

C13 driving segments $\square \quad b_k \leq DB; \quad \text{drive}_k - b_k \leq DB$

At most 4 h30 of driving before the break and after it. With no break ($b_k = 0$) this reads: total driving ≤ 4 h30.

C14 work segments $\square$

$a_k - (t_k^0 - P) \leq WB + M_k^w(1 - \text{brk}_k)$	work before the break
$(t_k^E + R) - (a_k + BR) \leq WB + M_k^w(1 - \text{brk}_k)$	work after the break
$(t_k^E + R) - (t_k^0 - P) \leq WB + M_k^w \text{brk}_k$	no break: whole duty

At most 6 h of work without a break. Prep P is work before the first segment; close R is work at the end of the last one.

Why this form. Leaving R out accepts plans that are illegal by up to 10 minutes. It was spotted by derivation while preparing the tests; the brute-force comparison of §9 then confirms it is detectable (1 instance in 15).

C15 daily driving $\square \quad \text{drive}_k \leq DD$

6.7 Service time and the week

C16 service time $\square \quad \theta_k \geq (t_k^E + R) - (t_k^0 - P) - BR \text{brk}_k - M_k^w(1 - \text{used}_k), \quad \theta_k \leq DS$

Today's *temps de service*, capped at 12 h. An idle driver has $\theta_k = 0$, so idle days do not consume weekly hours.

Why this form. θ_k is bounded from below only; it can be over-estimated but never under-estimated, which is the safe direction for every cap. Reported values and the next day's state are recomputed from the routes, never read from θ_k .

C17 weekly caps $\square \quad W_k + \theta_k \leq H_k^{\max}; \quad V_k + \text{drive}_k \leq \text{WD}$

C18 extra minutes $\square \quad o_k \geq W_k + \theta_k - H_k^{\text{thr}}, \quad o_k \geq 0$

The week enters the day here: the same route is cheaper for a driver at 20 h than for one at 37 h.

6.8 Optional: identical drivers

If drivers k and $k + 1$ are identical (contract, truck, shift, state), add $\sum_i \text{visit}_{ik} \geq \sum_i \text{visit}_{i,k+1}$ to remove mirror solutions. $\square$

6.9 A property worth knowing

Every day is feasible by construction. Postponing all orders and leaving all drivers idle satisfies every constraint, provided yesterday's plan was legal ($W_k \leq H_k^{\max}$) and $P + R \leq \text{WB}$. So an INFEASIBLE status can only mean inconsistent input data.

7 Which rule is enforced where

Rule	Source	Constraints
45 min break after at most 4 h30 of driving	Reg. (EC) 561/2006 art. 7	C10–C13
No more than 6 h of work without a break	Dir. 2002/15/EC art. 5; C. transports L3312-2	C10–C12, C14
Break ≥ 30 / 45 min if work 6–9 h / >9 h	same	C14 with A10
Daily driving ≤ 9 h	561/2006 art. 6	C15
Daily <i>temps de service</i> ≤ 12 h	C. transports D3312-51	C16
Weekly driving ≤ 56 h	561/2006 art. 6	C17
Weekly <i>temps de service</i> ≤ 52 h, extra pay above threshold	C. transports D3312-45, -50	C17, C18
Daily rest ≥ 11 h	561/2006 art. 8	A12 + week-loop check
Receiving windows	commercial	bounds of T , C5–C7
Capacity, access, tail-lift	physical	C4, sets C_k
Shift start and latest end	roster	bounds of t^0, t^E

8 Big-M values

Each big-M is the smallest value that switches its constraint off when the controlling binary is 0. Never use one global constant: loose big-M values are the main cause of slow MILPs of this type.

Where	Value	Reasoning
C5	$M_{jk}^0 = \max(0, F_k + \tau_{0j} - e_j)$	latest departure + travel, minus earliest possible T_j
C6	$M_{ij} = \max(0, l_i + s_i + \text{BR} + \tau_{ij} - e_j)$	latest end at i incl. break, minus earliest T_j
C7	$M_{ik}^E = \max(0, l_i + s_i + \text{BR} + \tau_{iE} - S_k - P)$	same, against the earliest return
C11	$M_{ik}^a = \max(F_k, l_i + s_i) - \min(S_k, e_i + s_i)$	full range of $a_k - (T_{ik} + s_i)$
C14, C16	$M_k^w = F_k - S_k + P + R$	longest possible duty
C8, C12, upper C9	DD	range of accumulated driving
lower C9	$\text{DD} + \tau_{ij}$	D_{ik} can reach DD on an unused arc

9 Evidence that the model is right

Before writing the brief, the formulation was implemented in a scratch script and compared against *exhaustive enumeration* (every assignment, sequence, postponement, break position and integer departure minute) on small days with 5 orders and 2 drivers, in three regimes.

Model variant	mixed (12)	work-heavy (15)	driving-heavy (15)
Formulation of this document	0 mismatches	0	0
Break delay removed from C6–C7	—	15 / 15 detected	—
C14 without R	0 / 12	1 / 15 detected	—
C9 lower side only	—	0 / 15	0 / 15
C9 lower big-M = DD	0 / 12	0 / 15	0 / 15

Random instances do not catch every error. The one-sided C9 is only exposed by a handcrafted case — the **trap**, which becomes a mandatory test:

Route depot $\rightarrow$ A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ depot, legs of 60, 140, 200 and 140 minutes; driving on arrival is 60 at A, 200 at B, 400 at C, and 540 in total.

Break after	Driving before	Driving after	4 h30 rule
A	60	480	violated
B	200	340	violated
C	400	140	violated
none	—	540	violated

No break position makes this route legal. The correct model postpones one order (cost 1 532.40 €); the model with a one-sided C9 serves all three illegally (cost 742.00 €).

The lower big-M of C9 was not caught by any test either; it is corrected by derivation (§8), not by evidence.

10 Method

10.1 One day

1. **Prepare.** Compatibility sets C_k ; arc pruning; big-M values.

2. **Baseline (status quo).** k-means territories, one per driver; Hungarian assignment of territories to drivers; nearest neighbour then 2-opt; the route evaluator chooses the break position and the departure time. Orders that fit nowhere are postponed.
3. **Solve.** Build the MILP of §6, give the baseline as a MIP start, solve with Gurobi (120 s, 1 % gap for *small*).
4. **Extract** routes, break positions and times.
5. **Check.** The independent checker recomputes every time from the plan and verifies every rule. Zero violations is mandatory.
6. **Fallback.** No incumbent at the time limit $\Rightarrow$ keep the baseline plan and flag the day.

10.2 The week

Algorithm 1: Rolling week (Monday to Friday)

```

1  $W_k \leftarrow 0, V_k \leftarrow 0$  for every driver;  $\text{carry} \leftarrow \emptyset$ ;
2 for  $d = \text{Monday to Friday}$  do
3    $C \leftarrow \text{orders of day } d \cup \text{carry}$ ; // carried orders keep their window and penalty
4    $\text{plan}_d \leftarrow \text{SOLVEDAY}(C, \text{drivers}, W, V)$ ; // steps 1-6 above
5   assert  $\text{checker}(\text{plan}_d) = \emptyset$ ;
6   foreach driver  $k$  do
7      $W_k += \theta_k, V_k += \text{drive}_k$ ; // recomputed from the route
8     check rest:  $\text{start}_{d+1} \geq \text{end}_d + 11 \text{ h}$ ;
9    $\text{carry} \leftarrow \text{postponed orders of } \text{plan}_d$ ;
10 report  $\text{carry}$  as unserved; compute weekly KPIs; run the week-mode checker;
```

Known limitation. The loop is *myopic*: Monday’s plan does not know Friday’s volume, so it may use a driver’s hours early and pay extra time later. Optional task T11 measures this *price of myopia* with a clairvoyant weekly MILP on small instances.

11 Size and complexity

Orders	Drivers	Variables	Constraints	Possible plans	Free pip licence
8	2	214	534	4×10^5	yes
13	3	697	1 852	6×10^{11}	yes (limit)
20	4	1 968	5 440	4×10^{21}	no
30	6	6 192	17 580	8×10^{37}	no
40	8	14 176	40 800	5×10^{55}	no

Sizes grow as $\text{drivers} \times \text{orders}^2$ (before arc pruning). Expected behaviour with a 2–5 minute limit: optimal up to about 15 orders a day, small gaps up to about 25, the wrong tool beyond about 40. Task T9 replaces these expectations with measurements.

12 Decisions that are yours

D1 — Break model

Keep one 45-minute break at a customer (A6, A7)? Alternative: allow the 15 + 30 split, i.e. two break slots. Doubles the break variables; closer to the law; rarely changes the plan in regional distribution.

D2 — Waiting counts as work (A9)

Conservative. Alternative: let waiting of known duration count as availability rather than work, which relaxes the 6 h rule.

D3 — Cost of regular hours (A14)

Tie-breaker only. Alternative: a full hourly cost, which turns the objective into “minimise hours” and favours shorter days even when nothing is saved.

D4 — Postponement penalty

Fixed 200€ per order, constant over the week. Alternative: escalate it for carried orders so a postponed order is not postponed twice.

D5 — Part-time ceiling

Contract hours + 10 % (Code du travail default). Alternative: up to one third, if a collective agreement allows it.

D6 — Rolling or clairvoyant as the headline

Rolling is realistic; clairvoyant is a bound. V0 reports rolling and, optionally, the gap to clairvoyant.

Sign-off. Assumptions A1–A16 ☐ Variables ☐ Constraints C1–C18 ☐ Decisions D1–D6 recorded ☐